

Unit

1

Lesson

2

Spark The Challenge!

Imagine, Sketch, Invent.

Time Frame: [80-90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Use the engineering design process to solve a communication problem.
- Brainstorm creative, inclusive tech ideas that help people who can't speak or hear.
- Identify the role and responsibilities of a Speech-Language Pathologist (SLP) and explain how they help people with communication challenges.
- Build essential reasoning skills.

Challenge questions of the lesson

- How can the Engineering Design Process help solve problems faced by people who can't speak or hear?
- What are some creative and inclusive technology ideas that could assist people with communication challenges?
- What does a Speech-Language Pathologist (SLP) do to help individuals with speech or hearing difficulties?
- How can you use reasoning skills to make better decisions during the design and problem-solving process?

SEL Relation

- **Promoting Empathy:** Encouraging students to understand and appreciate the challenges faced by people with communication disabilities.
- **Building Social Awareness:** Increasing awareness of diverse communication needs and inclusive technologies.
- **Enhancing Relationship Skills:** Facilitating collaboration, active listening, and respectful communication during group activities.
- **Fostering Responsible Decision-Making:** Guiding students to thoughtfully evaluate design ideas and make informed choices.

Engage



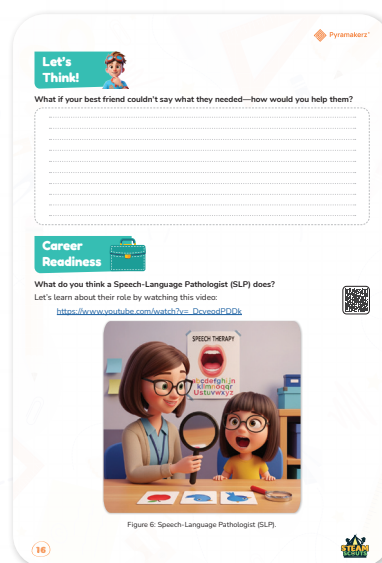
Teacher's Role:

- Read the story aloud with enthusiasm, highlighting the characters' curiosity and problem-solving mindset.
- Pause to check for understanding and invite students to imagine themselves in the scenario, thinking like engineers.
- Facilitate the "Let's Think" activity by asking the question aloud and encouraging students to reflect on how they would help a friend who can't speak.
- Provide examples or prompts if students need help starting their responses.
- Support diverse learners by clarifying vocabulary and encouraging all students to participate.

Expected from students:

- Listen actively and engage with the story, imagining themselves as part of the problem-solving team.
- Reflect thoughtfully on the "Let's Think" question and share ideas about helping someone who cannot speak.
- Participate respectfully in class discussions, listening to peers and building on their ideas.
- Express their thoughts clearly, either verbally or through writing.

Parts of the book included.



Story



Begin by gathering the class and setting a collaborative and curious tone.

- Say:
 - “Today, we’re going to think like engineers! We’ll see how Adam and Laila start solving a big challenge—helping someone who can’t speak share their thoughts. Let’s listen carefully to their plan.”
- Read the story aloud with expression and natural pacing, emphasizing excitement, curiosity, and teamwork.
- Pause occasionally to ask guiding questions such as:
 - “What problem are Adam and Laila trying to solve?”
 - “Why is it important to understand the problem before coming up with ideas?”
 - “What ideas did they suggest to help Samir communicate?”
- Encourage students to imagine themselves as part of the team, thinking about how they might help.
- Invite a few students to share quick thoughts or predictions before moving to the next part.

Possible Strategies for implementation:

1. Create a Collaborative Learning Environment.

- Start by encouraging students to listen actively and think like engineers working together.
- Use positive, inclusive language to build a safe space for sharing ideas.

2. Expressive and Interactive Reading.

- Use varied tone, volume, and pacing to highlight different characters and emotions.
- Pause at key moments to ask open-ended questions that prompt students to reflect and predict.

3. Use Guiding Questions to Scaffold Understanding.

- Ask questions before, during, and after reading to focus attention on the Engineering Design Process steps (e.g., identifying the problem, brainstorming solutions).
- Examples: “What problem are they trying to solve?”, “Why do you think brainstorming is important?”

4. Connect Story to Student Experiences.

- Relate Adam and Laila’s teamwork to students’ own group work or problem-solving experiences.
- Invite students to share times they worked with others to solve a challenge.

5. Visualize and Role-Play.

- Encourage students to imagine themselves as engineers in the story.
- Optionally, have volunteers act out parts to deepen engagement and understanding.

6. Clarify Vocabulary and Concepts.

- Explain any new terms (e.g., Engineering Design Process) in simple language.
- Use visual aids or anchor charts to reinforce key ideas.

7. Encourage Predictive and Reflective Thinking.

- Prompt students to guess what the next step in the process might be.
- o After reading, have them reflect on why the first steps of EDP are critical.

8. Support Diverse Learners.

- Offer sentence starters or prompts for students who need language support.
- Use visuals or graphic organizers to help organize story events and ideas.

Let's Think!



Objective:

Students will develop empathy and problem-solving skills by reflecting on communication challenges and considering practical ways to support friends who cannot verbally express their needs.

Answer:

"If my best friend couldn't say what they needed, I would watch their facial expressions and body language carefully. I might ask yes/no questions or offer choices to help them communicate. I would also try to be patient and use pictures or gestures to understand what they want."

Possible Strategies for implementation:

1. Create a Reflective Environment.

- Encourage students to think quietly and personally about the question before sharing.
- Emphasize that there are no wrong answers and that all ideas are valuable.

2. Facilitate Whole-Class or Small Group Discussion.

- After individual reflection, invite students to share their thoughts in pairs or small groups to build confidence.
- Lead a whole-class discussion where diverse strategies and feelings are shared.

3. Use Guiding Prompts.

- Provide sentence starters such as "If my best friend couldn't say what they needed, I would..."

- Ask probing questions to deepen thinking, like “Why would that help?” or “How would you know what they need?”

4. Connect to Personal Experiences.

- Encourage students to relate the question to real-life situations they’ve encountered or observed.
- Foster empathy by discussing feelings involved in being unable to communicate.

5. Support Diverse Learners.

- Allow responses verbally, in writing, or through drawings depending on student preferences.
- Offer additional time or scaffolded language support for those who need it.

Career Readiness



Objective:

Students will explore the role of a Speech-Language Pathologist (SLP) and understand how this career helps people overcome communication challenges, building awareness of STEAM-related professions and the importance of collaboration in problem-solving.



Video link: https://www.youtube.com/watch?v=_DcveodPDDk

Answers:

What do you think a Speech-Language Pathologist (SLP) does?

- A Speech-Language Pathologist helps people who have trouble speaking or understanding language.
- They work with children and adults to improve communication skills.
- SLPs design exercises and use technology to assist people with speech or hearing difficulties.
- They support people with disabilities to express themselves better and communicate effectively.

Possible Strategies for Implementation:

1. Activate Prior Knowledge

- Begin by asking students what they already know or guess about the job of a Speech-Language Pathologist.
- Encourage students to share ideas openly without worrying about correctness.

2. Video Viewing Preparation

- Introduce the video by explaining it shows how SLPs help people with speech and communication difficulties.
- Set a purpose for watching, e.g., “Listen carefully to learn what a Speech-Language Pathologist does every day.”

3. Guided Note-Taking or Discussion.

- Provide space for students to write answers to the initial question before or after watching.
- Pause after the video to discuss key points as a class, clarifying any confusing terms.

4. Connect to Lesson Theme.

- Relate the SLP's role to the communication challenges explored in the lesson.
- Discuss how different STEAM careers contribute to designing solutions like the talking device.

5. Encourage Reflection

- Prompt students to think about whether they might enjoy such a career and why.
- Optionally, have students write a brief reflection or share in small groups.

Explain



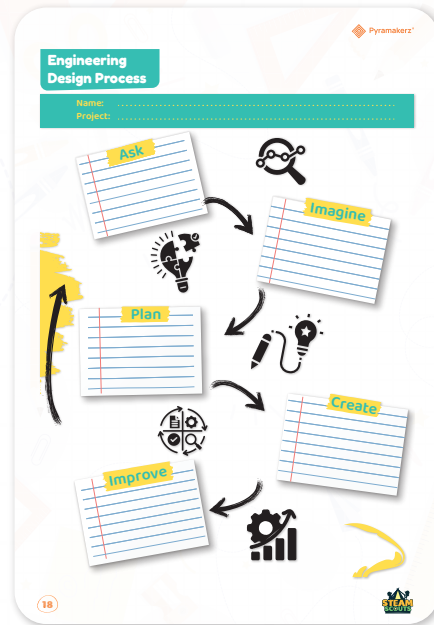
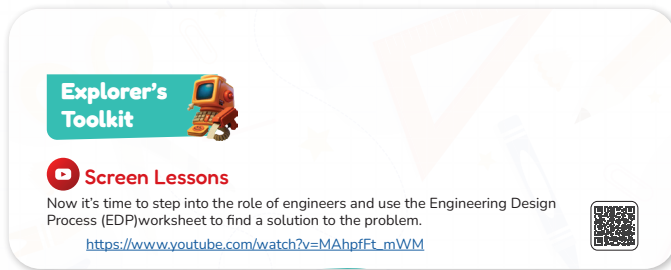
Teacher's Role:

- Prepare the classroom technology to ensure the video plays smoothly for all students.
- Introduce the video by explaining its focus on the Engineering Design Process (EDP) and how engineers use it to solve problems.
- Set a clear purpose for watching: students will learn the steps of the EDP and apply them to design solutions.
- Encourage attentive watching by reminding students to think about how each step helps solve problems.
- Pause the video at key moments (if possible) to highlight important concepts or clarify terms.
- Facilitate a brief discussion after the video to check comprehension and answer questions.
- Provide or guide students through the accompanying EDP worksheet to connect video concepts with hands-on practice.

Expected from students:

- Watch the video attentively, focusing on understanding each step of the Engineering Design Process.
- Take notes or mentally note key ideas to apply in the worksheet activity.
- Ask questions if any concepts are unclear.
- Complete the EDP worksheet thoughtfully, applying the process steps to the problem scenario.
- Participate in discussions by sharing insights or challenges about the EDP steps.

Parts of the book included.



Screen Lessons

Objective:

Students will understand the steps of the Engineering Design Process (EDP) and begin to think like engineers as they work to solve communication challenges by designing a talking device for people who have difficulty speaking.

Video link: https://www.youtube.com/watch?v=MAhpFt_mWM



Possible Strategies for Implementation:

1. Pre-Watching: Set the Purpose.

- Ask students:
 - “What do engineers do when they want to create something new?”
 - “Have you ever had to plan or design something before making it?”
- Tell students:
 - “In this video, we’ll learn how engineers follow steps to solve problems—just like we will when helping Samir communicate. Let’s see how to plan and build smart ideas!”

2. During Watching: Active Thinking.

- Pause at key moments to ask:
 - “Which step of the Engineering Design Process are they doing now—asking questions, imagining solutions, or planning?”
 - “What communication problem are they trying to solve for Samir?”
- Use a visual aid like an EDP chart or poster so students can point out or track the steps as they watch.

3. After Watching: Connect to Action.

- Discuss questions like:
 - “Which step of the process do you find most interesting?”
 - “What should be our first step to help Samir talk?”
- Guide students to begin filling in the ‘Ask’ and ‘Imagine’ sections of their EDP worksheet with ideas specific to Samir’s communication needs.

Engineering Design Process

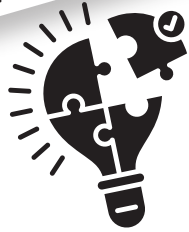
Name:

Project:

Ask

What is the problem?

Some individuals cannot speak or hear, making communication with others very difficult—especially with those who don't know sign language. How can we design a tool that turns hand movements into text to help them communicate clearly?



Imagine

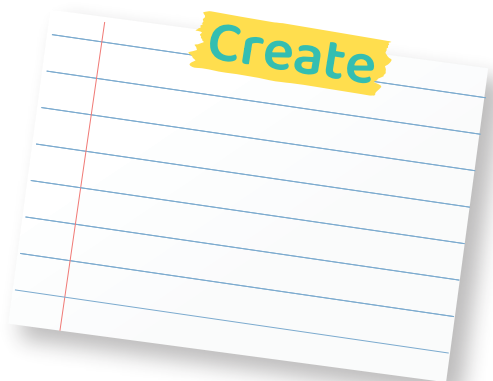
- A glove with sensors that send data to a microcontroller
- Converting finger bends to letters or phrases
- Displaying output on a screen or sending it to a phone

Plan

- Where sensors will go on the glove (fingers, palm, wrist)
- What kind of controller/display will be used (Arduino, LCD, mobile app)
- How the glove will detect gestures and convert them to readable text



Create



Improve



Elaborate and Evaluate



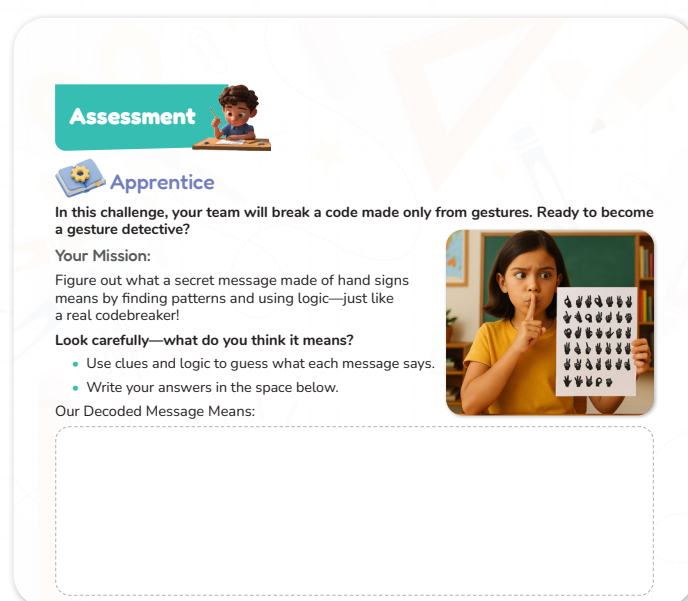
Teacher's Role:

- Introduce the activity by explaining the challenge: decoding a secret message made from gestures using clues and logic.
- Present the gesture codes clearly, ensuring all students understand the task and have access to materials.
- Encourage students to observe carefully, discuss ideas, and test hypotheses with their teammates.
- Monitor groups to provide guidance and prompt deeper thinking without giving away answers.
- Facilitate a whole-class discussion to review decoded messages and reasoning strategies.
- Support students who may struggle by offering hints or additional examples.

Expected from students:

- Work collaboratively in teams to analyze the gesture codes.
- Use logic and clues to infer the meaning behind each message.
- Write clear answers in the provided spaces.
- Share and explain their reasoning with peers during group discussions.
- Listen respectfully and consider alternative ideas.

Parts of the book included.



Apprentice

Objective:

Students will apply critical thinking and logical reasoning skills to decode messages made from hand gestures, practicing pattern recognition and problem-solving like real codebreakers.

Activity instructions:

1. Introduce the Challenge:

- Explain the mission to the class and emphasize the role of a “gesture detective.”

2. Review Gesture Codes:

- Show the coded hand gestures clearly, ensuring students understand the visual clues.

3. Teamwork and Brainstorming:

- Divide students into small teams.
- Encourage teams to discuss and hypothesize what each gesture might mean based on patterns or context.

4. Record Answers:

- Have students write their decoded meanings in the spaces provided on their worksheet.

5. Group Sharing:

- Invite teams to share their decoded messages and explain their reasoning process.
- Discuss similarities and differences in interpretations.

6. Reflection:

- Encourage students to reflect on how using logic and teamwork helped decode the message.

Example Coded Hand Gestures (Secret Message).

Below is a simple symbol-to-letter substitution based on unique hand gestures. Each hand pose stands for one letter. You can use images, emojis, or illustrated symbols to represent them, or just describe them like this:

Hand Gesture Description	Letter
One finger pointing up (index only)	H
Palm open with all five fingers spread	E
Thumb up (like a “thumbs up” sign)	L
Fist closed	P
Two fingers making a “peace” sign	M

Index and thumb make a circle (“OK” sign)	E
Hand in “stop” gesture (flat, palm out)	N
Three fingers up	O
Pinky finger raised only	W

Example Message for Students to Decode

- Gesture Sequence (students receive this):

1.    
2.  
3.   

Student Decoding Task (from above gestures):

1. H E L P
2. M E
3. N O W

Decoded Message: “HELP ME NOW”

Possible Strategies for Implementation:

1. Scaffold Critical Thinking

- Provide sentence starters or guiding questions, such as “This gesture might mean...” or “I noticed this pattern...”
- Model decoding one gesture together as a class before group work.

2. Encourage Collaboration

- Assign roles within teams (e.g., recorder, spokesperson) to ensure active participation.
- Foster respectful discussion and encourage students to build on each other’s ideas.

3. Use Visual Supports

- Display enlarged versions of the gesture codes for easy viewing.
- Provide graphic organizers to help organize clues and inferences.

4. Differentiate Support

- Offer additional hints or simplified codes for students who need extra help.
- Allow alternative presentation formats, like drawing or verbal explanations.

5. Connect to Real-World Skills.

- Highlight how logic, pattern recognition, and teamwork are used in many real-life careers, including coding, cryptography, and communication.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist.

- Read each statement aloud one at a time.
- After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
- Example prompts:
 - "Did you learn how people share ideas without talking?"
 - "Do you know how tools can help people talk?"

3. Student Completion

- Let students check off or color the "Now I Can" boxes they feel good about.
- Remind them: "It's okay if you didn't check all of them. We're all still learning."

4. Optional Extension

- Ask students to circle the one they're most proud of.
- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. NGSS (Next Generation Science Standards).

- **MS-ETS1-1:** Define problems that can be solved through engineering design.
- **MS-ETS1-2:** Evaluate competing design solutions based on criteria and constraints.
- **MS-LS1-8:** Gather information on how the body's sensory systems detect and process information.

2. CCSS (Common Core State Standards).

- **CCSS.ELA-LITERACY.SL.6.1:** Engage effectively in collaborative discussions.
- **CCSS.ELA-LITERACY.RI.6.3:** Analyze interactions between individuals, events, ideas in texts.

3. ISTE (International Society for Technology in Education).

- **Empowered Learner:** Students use technology to research and solve problems.
- **Creative Communicator:** Students communicate complex ideas clearly and creatively.